CHiMaD

Center for Hierarchical Materials Design

PHASE FIELD METHODS V

Upload-a-Thon / Workshop

NORTHWESTERN UNIVERSITY

CHIMAD.NORTHWESTERN.EDU

August 1st- 3rd, 2017

CHiMaD | Center for Hierarchical Materials Design

Center for Hierarchical Materials Design (CHiMaD) is a NIST-sponsored center of excellence for advanced materials research focusing on developing the next generation of computational tools, databases and experimental techniques in order to enable the accelerated design of novel materials and their integration to industry, one of the primary goals of the U.S. administration's Materials Genome Initiative (MGI).

This Chicago-based consortium includes Northwestern University as the lead, University of Chicago, Northwestern-Argonne Institute for Science and Engineering, a partnership between Northwestern University and Argonne National Laboratory, and the Computational Institute, a partnership between University of Chicago and Argonne National Laboratory. The consortium is also partnered with Questek Innovations, a pioneering materials design company, ASM International and Fayetteville State University.

Designing novel materials of specific properties for a particular application requires simultaneously utilizing physical theory, advanced computational methods and models, materials properties databases and complex calculations. This approach stands in contrast to the traditional trial-and-error method of materials discovery. CHiMaD aims to focus this approach on the creation of novel hierarchical materials which exploit distinct structural details at various scales, from the atomic on up, to obtain enhanced properties. The center's research focuses on both organic and inorganic advanced materials in fields as diverse as self-assembled biomaterials, smart materials for self-assembled circuit designs, organic photovoltaic materials, advanced ceramics and metal alloys.

For more information, please visit chimad.northwestern.edu

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Agenda
Phase Field Methods V
Upload-a-Thon / Workshop
August 1st-3rd, 2017

Tuesday, August 1st, 2017
CHiMaD Headquarters

11:45 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
12:00 PM	Kick-off lunch
1:00 PM	**Welcome** *Begum Gulsoy*
1:05 PM	**State of the CHiMaD Website** *Daniel Wheeler*
1:25 PM	**Upload-a-Thon**
4:00 PM	**Presentations of Upload-a-Thon results and discussion**
5:30 PM	Adjourn | *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Wednesday, August 2nd, 2017
CHiMaD Headquarters

8:30 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
8:45 AM	Breakfast
9:00 AM	**Presentations of Benchmark III results**
10:00 AM	Coffee break
10:15 AM	**Discussion of Benchmark III problems**
11:15 AM	**Presentation of current status of benchmark problems, the website, etc.**
12:30 PM	Lunch
1:30 PM	**PyMKS introduction** *Surya Kalidindi*

2:00 PM	**CALPHAD integration into phase field models** *Michael Welland*
2:30 PM	**Issues with adaptive meshing** *Nana Ofori Opoku*
3:00 PM	Coffee break
3:15 PM	**Uncertainty quantification in phase field modeling** *Steve DeWitt*
3:45 PM	**Outstanding problems in phase field modeling** *Peter Voorhees*
4:15 PM	**Open discussion**
5:00 PM	Reception
7:30 PM	Adjourn | *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Thursday, August 3rd, 2017

8:30 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
8:45 AM	Breakfast
9:00 AM	**Tutorial on pyMKS** *Daniel Wheeler*
9:30 AM	**PyMKS lab session**
10:45 AM	Coffee break
11:00 AM	**Phase field simulations of irradiation effects** *Marius Stan*
11:30 AM	**Wrap-up discussion**
12:00 PM	Adjourn (lunches to-go) *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Logistics
Phase Field Workshop
August 1-3, 2017

Workshop Location

CHiMaD Headquarters
Northwestern University
2205 Tech Drive
Evanston, IL 60208

Parking Information

Please use the *North Campus Parking Garage* for parking on campus.

If driving, please park at the North Garage, and take a ticket as you enter. Validation tickets will be available at CHiMaD Headquarters.

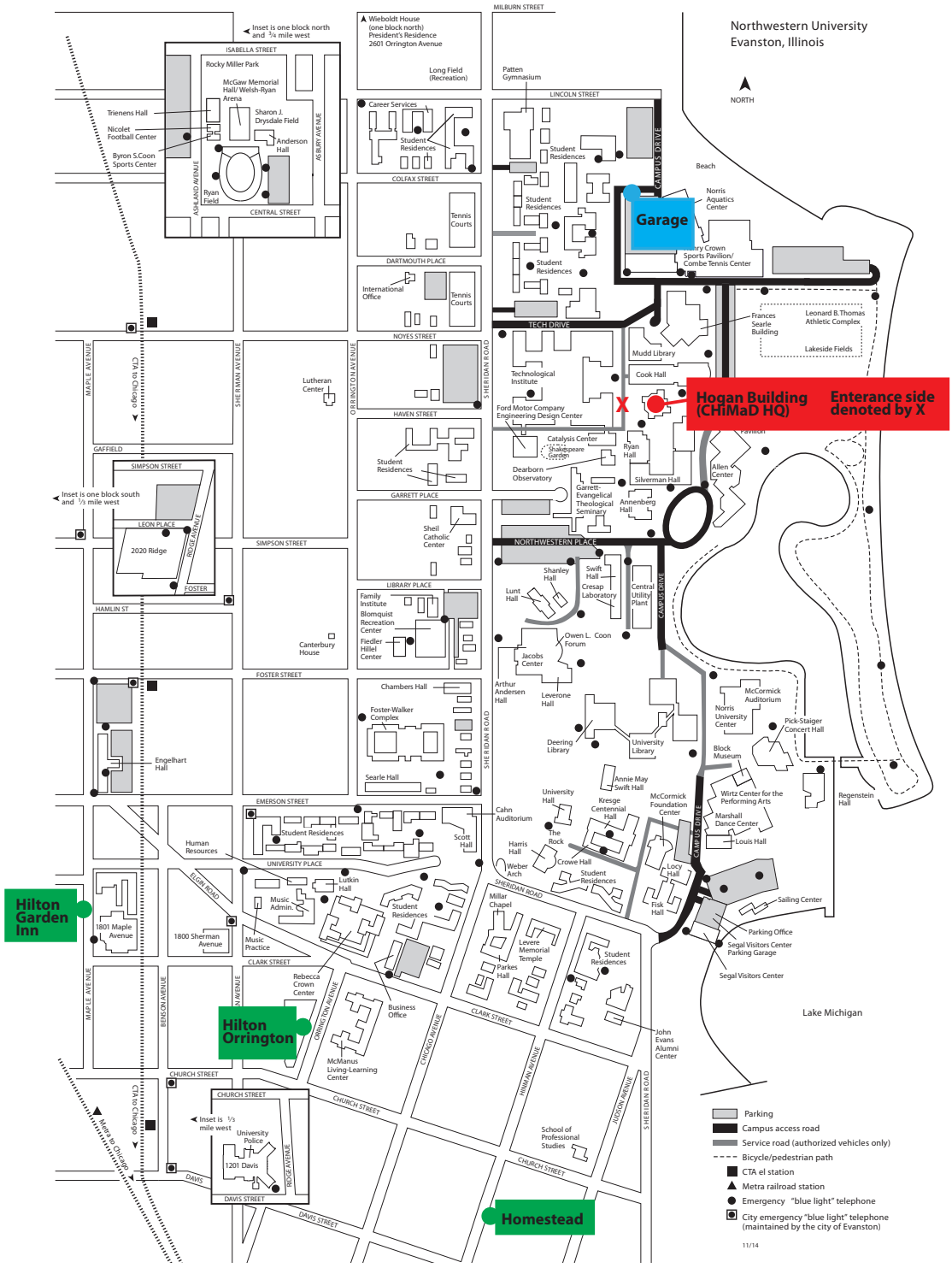
Hotel & Shuttle Information

Hilton Garden Inn
1818 Maple Ave
Evanston, IL 60201
847-475-6400

A shuttle is available to and from the workshop, but you are also welcome to walk if you prefer. Please see the map on the next page for locations.

Reimbursements

Please contact *Victoria Fonseca*, victoria.fonseca@northwestern.edu for details.


Northwestern University
Evanston, Illinois
NORTH
Garage
Hogan Building
(CHiMaD HQ)
Enterance side
denoted by X
X
Hilton
Garden
Inn
Hilton
Orrington
Homestead
Lake Michigan
Inset is one block north
and ¾ mile west
Inset is one block south
and ⅓ mile west
Inset is ⅓
mile west
Wieboldt House
(one block north)
President's Residence
2601 Orrington Avenue
Long Field
(Recreation)
ISABELLA STREET
Rocky Miller Park
McGaw Memorial
Hall/ Welsh-Ryan
Arena
Trienens Hall
Nicolet
Football Center
Byron S.Coon
Sports Center
Sharon J.
Drysdale Field
Anderson
Hall
Ryan
Field
ASHLAND AVENUE
ASBURY AVENUE
CENTRAL STREET
Career Services
Student
Residences
COLFAX STREET
Tennis
Courts
DARTMOUTH PLACE
International
Office
NOYES STREET
MAPLE AVENUE
CTA to Chicago
SHERMAN AVENUE
ORRINGTON AVENUE
SHERIDAN ROAD
Lutheran
Center
HAVEN STREET
GAFFIELD
SIMPSON STREET
LEON PLACE
2020 Ridge
RIDGE AVENUE
FOSTER
GARRETT PLACE
Sheil
Catholic
Center
LIBRARY PLACE
Family
Institute
Blomquist
Recreation
Center
Fiedler
Hillel
Center
HAMLIN ST
Canterbury
House
FOSTER STREET
Chambers Hall
Foster-Walker
Complex
Searle Hall
Engelhart
Hall
EMERSON STREET
Student Residences
Scott
Hall
Human
Resources
UNIVERSITY PLACE
Lutkin
Hall
Music
Admin.
Student
Residences
EVON ROAD
1801 Maple
Avenue
1800 Sherman
Avenue
Music
Practice
CLARK STREET
Rebecca
Crown
Center
Business
Office
McManus
Living-Learning
Center
CHICAGO AVENUE
CHURCH STREET
DAVIS
DAVIS STREET
University
Police
1201 Davis
Metra to Chicago
MILBURN STREET
Patten
Gymnasium
LINCOLN STREET
Student
Residences
CAMPUS DRIVE
Beach
Norris
Aquatics
Center
Henry Crown
Sports Pavilion/
Combe Tennis Center
TECH DRIVE
Frances
Searle
Building
Leonard B.Thomas
Athletic Complex
Lakeside Fields
Mudd Library
Cook Hall
Technological
Institute
Ford Motor Company
Engineering Design Center
Catalysis Center
Shakespeare
Garden
Ryan
Hall
Dearborn
Observatory
Silverman Hall
Allen
Center
Garrett-
Evangelical
Theological
Seminary
Annenberg
Hall
NORTHWESTERN PLACE
Shanley
Hall
Swift
Hall
Lunt
Hall
Cresap
Laboratory
Central
Utility
Plant
Owen L. Coon
Forum
Jacobs
Center
Arthur
Andersen
Hall
Leverone
Hall
Deering
Library
University
Library
McCormick
Auditorium
Norris
University
Center
Pick-Staiger
Concert Hall
Block
Museum
Annie May
Swift Hall
University
Hall
Kresge
Centennial
Hall
McCormick
Foundation
Center
Wirtz Center for the
Performing Arts
Marshall
Dance Center
Louis Hall
Regenstein
Hall
Cahn
Auditorium
The
Rock
Harris
Hall
Crowe Hall
Weber
Arch
Student
Residences
Locy
Hall
Fisk
Hall
Sailing Center
Parking Office
Segal Visitors Center
Parking Garage
Segal Visitors Center
SHERIDAN ROAD
Millar
Chapel
Levere
Memorial
Temple
Parkes
Hall
Student
Residences
John
Evans
Alumni
Center
HINMAN AVENUE
HOGAN AVENUE
School of
Professional
Studies
Parking
Campus access road
Service road (authorized vehicles only)
Bicycle/pedestrian path
CTA el station
Metra railroad station
Emergency "blue light" telephone
City emergency "blue light" telephone
(maintained by the city of Evanston)
11/14